SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: DSA0204**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage:20%

Code of Conduct:

I __Chalamala Sai Harshitha (192424346)__ *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Sai Harshitha

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Depth Estimation Strategy

Constraints:

- **Only one camera available**
- **Moderate depth accuracy acceptable**

Options:

Stereo Vision: high accuracy, needs two cameras

Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

Factor	Stereo Vision	Structure from Motion (SfM)
Depth accuracy	High	Moderate
Camera requirement	Two cameras	One camera
Hardware requirement	High	Low
Suitable for given system	No	Yes
Complexity	Higher	Lower
Accuracy requirement	More than required	Sufficient

Analysis of Stereo Vision

Stereo Vision estimates depth using two cameras from different viewpoints.

Advantages:

- Provides high depth accuracy.
- Produces reliable depth information when cameras are properly calibrated.
- Suitable for applications where precise depth measurement is required.

Limitations:

- Requires **two cameras**.
- The given system has only one camera available.
- Additional hardware would increase cost and system complexity.

Therefore, although Stereo Vision provides better accuracy, it cannot directly satisfy the hardware constraint.

Analysis of Structure from Motion

Structure from Motion estimates 3D information from multiple images captured using a single moving camera.

Advantages:

- Works with one camera.
- Provides moderate depth accuracy.
- Matches the given requirement because moderate accuracy is acceptable.
- Requires less hardware than Stereo Vision.
- More practical for systems with limited camera resources.

Limitation:

- Its depth accuracy is lower than Stereo Vision.
- It depends on sufficient camera or scene motion to obtain useful depth information.

Since the requirement only specifies moderate depth accuracy, this limitation is acceptable.

Constraint-Based Evaluation

The most important constraints are camera availability and required accuracy.

- **Camera constraint:** SfM satisfies the requirement because it uses one camera, while Stereo Vision does not.
- **Accuracy constraint:** SfM provides moderate accuracy, which is sufficient for the stated requirement.
- **Hardware constraint:** SfM requires less hardware and is therefore more feasible.
- **Practical deployment:** SfM is easier to implement when adding a second camera is not possible.

Final Decision

Structure from Motion (SfM) should be selected.

Although Stereo Vision provides higher accuracy, it requires two cameras, which directly violates the given hardware constraint. SfM provides the required moderate accuracy while operating with the available single camera.

Conclusion

Based on the given constraints, SfM is the most feasible depth estimation approach. Stereo Vision offers higher accuracy but cannot satisfy the one-camera requirement. Since moderate accuracy is acceptable, SfM provides the better balance between accuracy, hardware availability, and practical deployment. Therefore, Structure from Motion should be chosen for this system.

2. Retail Motion Tracking Logic

A retail system requires:

- **Accurate tracking when crowd density is low**
- **Stable tracking when density is high**

Observations:

Optical Flow: accurate in low density, noisy in high density

Motion Models: stable in high density, less detailed

Compare both and evaluate a suitable approach or combination. Justify your reasoning logically.

Factor	**Optical Flow**	**Motion Models**
Low-density tracking	Accurate	Less detailed
High-density tracking	Noisy	Stable
Detail level	High	Moderate
Stability	Lower in crowds	High in crowds
Best condition	Low crowd density	High crowd density

Analysis

Optical Flow tracks the movement of pixels between consecutive frames.

- Provides **accurate and detailed tracking** when fewer people are present.
- Can identify individual movements effectively in low-density areas.
- In crowded scenes, movements from multiple people overlap.
- This produces noisy motion vectors and reduces tracking stability.

Motion Models use a model of expected movement to maintain stable tracking.

- Performs better when **crowd density is high**.
- Provides stable tracking even when individual motion details are difficult to distinguish.
- However, it is less detailed than Optical Flow.
- Therefore, it may not provide the same level of accuracy in sparse scenes.

Evaluation

Neither technique is ideal for **all crowd conditions**.

When the crowd is **low-density**, Optical Flow should be preferred because its detailed motion information provides more accurate tracking.

When the crowd becomes **high-density**, Motion Models are more suitable because stability becomes more important than detailed individual motion.

Recommended Approach

A **hybrid or adaptive approach** would provide the best solution:

- **Low crowd density → Optical Flow**
- **High crowd density → Motion Models**

The system can first estimate crowd density and then select the appropriate tracking method. This allows the system to take advantage of the strengths of both approaches.

Advantages of the Combined Approach

- Better accuracy in low-density areas.
- Improved stability in crowded areas.
- Reduces the noise problem associated with Optical Flow.
- Maintains useful tracking when individual motion becomes difficult to distinguish.
- Provides a more flexible solution for real-world retail environments.

Conclusion

A combination of Optical Flow and Motion Models is the most suitable approach. Optical Flow should be used when the crowd is sparse because it provides accurate and detailed tracking. Motion Models should be used when the crowd becomes dense because they provide greater stability. Therefore, an adaptive system that switches between the two techniques according to crowd density would provide a better balance between accuracy and stability.

3. Defect Detection System Decision

Requirements:

- **Detect subtle defects (>95% accuracy)**
- **Adapt to product variations**

Options:

Traditional methods: fast, less adaptive

Deep learning: accurate, requires training

Compare both approaches and evaluate which meets requirements. Justify using performance vs adaptability logic.

Factor	Traditional Methods	Deep Learning
Accuracy	Lower	High
Defect detection	Less effective for subtle defects	Better for subtle defects
Adaptability	Low	High
Product variations	Less suitable	More suitable
Processing speed	Fast	Higher computational requirement
Training requirement	Low	Requires training
Requirement suitability	Limited	High

Analysis

Traditional Computer Vision Methods

Traditional methods generally depend on manually designed features, thresholds, edges, shapes, or textures.

Advantages:

- Fast processing.
- Lower computational requirements.
- Easier to implement for simple and consistent defects.

Limitations:

- Less adaptive to changes in product appearance.
- May fail when defects are very small or subtle.
- Requires manual selection and adjustment of features or thresholds.
- Difficult to maintain when products have different shapes, textures, or surface patterns.

Deep Learning

Deep Learning learns important visual features directly from training images. This makes it more capable of identifying complex and subtle defect patterns.

Advantages:

- Provides higher detection accuracy.
- Better suited for subtle and complex defects.
- Can learn variations in product appearance.
- More adaptable when trained using diverse product images.
- Reduces dependence on manually designed features.

Limitation:

- Requires a sufficiently large and representative training dataset.
- Training requires more computational resources.
- Model performance depends on the quality and diversity of training data.

Requirement-Based Evaluation

The most important requirement is accuracy above 95%. Since traditional methods are described as less accurate, they may not reliably satisfy this requirement.

The second requirement is adaptation to product variations. Traditional methods are less adaptive, whereas Deep Learning can learn different product characteristics from training data.

Therefore:

- Accuracy requirement → Deep Learning
- Adaptability requirement → Deep Learning
- Fast processing → Traditional methods
- Overall requirement satisfaction → Deep Learning

Recommended Solution

Deep Learning should be selected because the system prioritizes subtle defect detection with more than 95% accuracy and adaptability to product variations.

To make the solution practical, the training dataset should contain images representing different product types, surface conditions, defect sizes, and variations. This helps the model generalize better to new products.

Conclusion

Therefore, Deep Learning is the most appropriate approach because it provides the required balance of high detection accuracy and adaptability, which are the primary requirements of the defect detection system.

4. AR Motion Tracking Constraint Problem

An AR system must:

- **Maintain stable tracking (<5% drift)**
- **Run on mobile hardware**

Options:

Optical Flow: detailed but computationally intensive

Motion Estimation: less detailed but efficient

Compare and evaluate which method satisfies both constraints. Justify your decision logically.

Factor	Optical Flow	Motion Estimation
Tracking detail	High	Lower
Computational requirement	High	Low
Mobile suitability	Lower	High
Tracking stability	Detailed tracking	Efficient tracking
Power/resource usage	Higher	Lower
Real-time suitability	Limited	Better

Analysis of Optical Flow

Optical Flow estimates the movement of pixels between consecutive frames and provides detailed motion information.

Advantages:

- Captures detailed motion information.
- Can provide accurate tracking of image features.
- Useful when fine motion details are important.

Limitations:

- Computationally intensive.
- Requires more processing resources.
- Can place a higher load on mobile hardware.
- Higher computation can affect real-time AR performance and resource usage.

Therefore, although Optical Flow provides detailed tracking, its computational requirement makes it less suitable for the given mobile constraint.

Analysis of Motion Estimation

Motion Estimation provides a less detailed representation of movement but is more computationally efficient.

Advantages:

- Requires fewer computational resources.
- Better suited for mobile hardware.
- Supports faster processing.
- Reduces the computational burden on the device.
- More practical for real-time AR applications.

Limitation:

- Provides less detailed motion information compared with Optical Flow.

However, the question prioritizes stable tracking and mobile deployment, rather than maximum motion detail.

Constraint-Based Evaluation

The two main constraints are:

1. Stable tracking (<5% drift):

The selected method must maintain sufficiently stable tracking. Motion Estimation is designed as the more efficient option and is therefore better suited to maintaining continuous tracking on a resource-limited device.

2. Mobile hardware:

Motion Estimation is clearly more suitable because it requires less computation. Optical Flow may impose excessive computational load on mobile hardware.

Recommended Approach

Motion Estimation should be selected for the given AR system.

Although Optical Flow provides more detailed motion information, its computational intensity conflicts with the mobile hardware constraint. Motion Estimation provides a better balance between tracking stability, computational efficiency, and mobile deployment.

If additional processing resources are available, Optical Flow could be used selectively for situations where detailed motion information is necessary.

Conclusion

Therefore, Motion Estimation provides the better constraint-based solution for the given AR application.

5. Mobile Face Detection Trade-off

Constraints:

- **Real-time operation required**
- **Accuracy ≥85%**

Options:

Viola-Jones: 80%, fast

Deep Learning: 95%, slow

Compare and evaluate which approach should be chosen. Justify using constraint satisfaction logic.

Factor	**Viola-Jones**	**Deep Learning**
Accuracy	80%	95%
Processing speed	Fast	Slow
Accuracy requirement	85% or higher	85% or higher
Real-time requirement	Suitable	Less suitable
Mobile deployment	More suitable	More demanding
Main advantage	Fast detection	High accuracy
Main limitation	Lower accuracy	Higher processing requirement

Analysis of Viola-Jones

Viola-Jones is a traditional face detection technique designed for fast processing. Its low computational requirement makes it suitable for real-time applications and mobile devices.

However, its accuracy is only 80%, which is below the required 85%. Therefore, although it satisfies the speed requirement, it does not satisfy the minimum accuracy requirement.

Analysis of Deep Learning

Deep Learning provides an accuracy of 95%, which is above the required 85%. It can learn complex facial patterns and therefore provides better detection performance.

However, the method is described as slow. This creates a problem because the system must operate in real time. Its higher computational requirement can also be challenging for mobile hardware.

Therefore, Deep Learning satisfies the accuracy requirement but creates a difficulty with real-time operation.

Constraint-Based Evaluation

The system has two mandatory requirements:

1. **Accuracy must be at least 85%.**
2. **The system must operate in real time.**

Based on the given data:

Requirement	**Viola-Jones**	**Deep Learning**
Accuracy ≥ 85%	80%	95%
Real-time operation	Fast	Slow
Overall suitability	Limited	Limited

Neither of the two given approaches completely satisfies both constraints.

Recommended Solution

Since the system requires both high accuracy and real-time performance, directly selecting either method would involve violating one of the requirements.

A practical solution would be to use a lightweight or optimized Deep Learning model designed for mobile deployment. This can retain the accuracy advantage of Deep Learning while reducing its computational cost and processing time.

If the system must strictly choose between the two given approaches, Viola-Jones can be considered when real-time response is the highest priority, but its accuracy must be improved to meet the 85% requirement.

Trade-Off

The problem represents a clear accuracy versus speed trade-off:

- **Viola-Jones:** Faster processing but insufficient accuracy.
- **Deep Learning:** Higher accuracy but slower processing.
- **Optimized Deep Learning:** Potentially provides a better balance between both requirements.

Conclusion

Therefore, the most suitable practical solution is an optimized lightweight Deep Learning approach that can achieve at least 85% accuracy while maintaining real-time performance on mobile hardware. This provides the best balance between the required accuracy, speed, and deployment constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation